SOUTH ASIA EDITION

Cohen's PATHWAYS of the PULP

Twelfth Edition

Web Editor:
ILAN ROTSTEIN

LOUIS H. BERMAN
KENNETH M. HARGREAVES

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South Asia Edition

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Dr. Hargreaves is Professor and Chair of the Department of Endodontics at the University of Texas Health Science Center at San Antonio. He is a Diplomate of the American Board of Endodontics and maintains a private practice limited to endodontics. He is an active researcher, lecturer, and teacher and serves as the Editor-in-Chief of the *Journal of Endodontics.* He is principal investigator on several nationally funded grants that combine his interests in pain, pharmacology, and regenerative endodontics. He has received several awards, including a National Institutes of Health MERIT Award for pain research, the AAE Louis I. Grossman Award for cumulative publication of research studies, and two IADR Distinguished Scientist Awards.



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He has served in leadership roles for various dental organizations, including Chair of the International Federation of Endodontic Associations' Research Committee; as a committee member of the American Association of Endodontists and European Society of Endodontology; and as a scientific reviewer for international endodontic and dental journals. He has also served as President of the Southern California Academy of Endodontists, Israel Endodontic Society, and International Association for Dental Research—Israel Division and as Chair of the Israel National Board of Diplomates in Endodontics.

Dr. Rotstein has published more than 150 scientific papers and research abstracts in the dental literature as well as chapters in international endodontic textbooks, including *Pathways of the Pulp*, *Ingle's Endodontics*, *Endodontics: Principles and Practice*, *Seltzer and Bender's Dental Pulp*, and *Harty's Endodontics in Clinical Practice*. He has lectured extensively in more than 25 countries throughout 5 continents.

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Dr. John Ingle

The development of every edition of *Cohen's Pathways of the Pulp,* for all of its editors and contributors, is a journey into both the future and the past of endodontics. What we as clinicians know today and the care that knowledge enables us to provide to our patients are the result of the curiosity, dedication, and commitment of the teachers, researchers, and clinicians who have come before us. Dr. John Ingle, who contributed to the science, practice, and teaching of endodontics for more than 7 decades, is one of those extraordinary pioneers.

Dr. Ingle began his career as an educator at the University of Washington in Seattle, where he taught periodontics. While teaching, he became interested in the new field of endodontics and, perhaps drawing on the example of his pioneer great-grandfather, Daniel Boone, he entered a specialty program in endodontics at the University of Michigan to explore this new discipline, earning graduate degrees in endodontics and periodontics.

As an educator and clinician, Dr. Ingle knew the importance of evidence in establishing the efficacy of endodontic treatment as a new specialty. To meet this need, he evaluated the results of endodontic treatment in 3000 patients and presented his findings to the annual session of the American Association of Endodontists in 1953. The *Washington Study* established proof of outcome for endodontic treatment and remains a seminal work in the literature for our field.

New fields offer many new challenges to their pioneers, and Dr. Ingle soon turned his attention to the development of standardization of endodontic instruments. His work resulted in the metric measurements and the 0.2 taper of endodontics files that were established in 1957.

In 1965, Dr. Ingle brought together his experience as both a clinician and an educator in the publication of his foundational textbook *Ingle's Endodontics.* Now in its sixth edition, *Ingle's Endodontics* has provided an essential evidence-based reference to students and clinicians for more than 50 years.

While Dr. Ingle's many accomplishments are known and admired, the attributes of his character are equally revered. Dr. Ingle was modest about his many contributions to endodontics and was unfailingly generous in his support and encouragement of others. The compassion and empathy that motivated his work were experienced by all those he touched throughout his long and exemplary career and life.

It is with profound gratitude and appreciation that we dedicate this twelfth edition of *Cohen's Pathways of the Pulp* to Dr. John Ingle, a scholar, a leader, and a gentleman.

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Foreword

'Perfection is a myth' – Perfection is not attainable but if you chase perfection you can catch excellence-and it is truly an honour to pen down the introductory words for a prestigious compilation and work of excellence: "Cohen's Pathways of the Pulp", twelfth edition, authored by one of the greatest mentors Dr. Stephen Cohen, known for his breakthrough in clinical Endodontics. I feel privileged to write the foreword for the South Asian edition. As a student I revered, the one my mentors referred and the one I further advocate to my students.

Since its introduction in 1976, the book, widely read by clinicians and academicians alike, has undergone several revisions. Through the successive editions, the authors have been scrupulous in presenting the changing trends, paradigm shifts and evolving ideas encompassing minute details of the subject, right from core basics to the advent in technology, original research and evidence based statements, thereby presenting the entire scope, capacity and the milestones of our specialty. The present edition provides an update on, 'Lesions that mimic Endodontic Pathosis', 'Managing Iatrogenic Events', 'Root Resorption' coupled with reorganising the chapters into three parts: The Core Science of Endodontics, Advanced Science Topics, and Advanced Clinical Topics, for the benefit of the reader; which make this book indispensable in its use and reference.

I profoundly believe that this edition will provide a wide variety of facts to enhance and achieve in-depth and sound knowledge of the Endodontic discipline that will provide incentive and impetus for further reading and refinement of the diagnostic, therapeutic and prognostic aspects of Endodontics.

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New to This Edition

The radiographic interpretation of odontogenic and non-odontogenic lesions is exactly that: an "interpretation." This new twelfth edition boasts a completely new chapter entitled Lesions That Mimic Endodontic Pathosis that elucidates and differentiates lesions that may appear as endodontic origin. This is a perfect adjunct to the chapters on Diagnosis and Radiographic Interpretation.

The chapter on Managing Iatrogenic Events has been completely rewritten to include an expansive section on injury to the inferior alveolar nerve.

Damage to the inferior alveolar nerve secondary to endodontic treatment is an avoidable dilemma. There is now specific content elaborating on the avoidance and management of these types of injuries.

Root resorption and root fractures can be some of the most difficult defects to clinically manage. The Root Resorption chapter on these subjects has been completely updated and will prove beneficial to the clinician and academician.

This edition updates all of the previous chapters to reflect the changes in the literature since the last edition.

New Chapter Organization

Chapters have been reorganized and grouped into three parts: Part I: The Core Science of Endodontics, Part II: Advanced Science Topics, and Part III: Advanced Clinical Topics. The twelve chapters in Part 1 focus on the core clinical concepts for dental students, while the chapters in Parts II and III provide the information that advanced students and endodontic residents and clinicians need to know.

The new organization better reflects the chronology of endodontic treatment.

Introduction

The foundation of the specialty of endodontics is a gift from the generations of great endodontists and researchers before us. They guided us with the goals of treatment, the benefits of their advancements, and the frailties of their deficiencies. From volumes of research, we have collectively built a virtual library of knowledge that leads us to the evidence we need for mastering our clinical procedures and benefiting our patients. As we look into our future, we should be directed toward developing the necessary tools for maximizing our outcomes with consistency, longevity, and, above all, patient well-being.

Over the past several decades, we have gone from arsenic to sodium hypochlorite, from bird droppings to gutta-percha, from hand files to motor-driven files, from culturing to one-visit appointments, from 2D to 3D radiography, and from pulp removal to pulpal regeneration. Yet still, the clinical and academic controversies are pervasive.

With patients living longer and with the inescapable comparison of endodontics to endosseous implants, the demand for endodontic excellence has greatly increased. Surprisingly, we still base our diagnosis on a presumed and almost subjective pulpal status. Imagine a future in which endodontic diagnosis could be made more objective by noninvasively scanning the pulp tissue. Imagine algorithms built into all digital radiography for interpreting and extrapolating disease processes. CBCT has made a huge impact on endodontic diagnosis, but can we enhance these digital captures with a resolution that would approach microcomputed tomography, and with less radiation? Will these 3D scans guide us not just with diagnostic objectivity, but also with direct treatment facilitation to guide us during surgical and nonsurgical treatment? Truly, we are now on the cusp of gaining the knowledge and technology for accomplishing this. As for clinical visualization, will 3D visualization and monitor-based observation change the way we visualize and implement our procedures? Will our procedures still be done with the fine motor skills of the clinician, or with the augmented reality of digital microsurgical devices? In the years to come, will we be able to truly eliminate all of the canal microorganisms, biofilms, and pulpal tissue? Will we be facilitating our canal cleaning with less toxic and more directed irrigants? Once we are finally able to totally clean and disinfect the canals to a microscopic level, will we have an obturation material that finally satisfies ALL the material requirements that Dr. Louis Grossman enumerated at the inception of our specialty? Will this obturating material be newly regenerated vital pulp?

Clearly, our endodontic future lies in out-of-the-box thinking, with the next generation of transformations coming with collaborations not just from within the biological sciences, but also in conjunction with physicists, chemists, engineers, and a multitude of other great innovative minds. The predictability of endodontics must be incontestable, not just with better technology to guide us toward greater success, but also to better elucidate exactly when endodontic treatment cannot be successful. Our future needs to focus on predictability and consistency, which will only be achieved with disruptive technologies, rather than persisting with variations and modifications of our current convictions. As a specialty, we have advanced by leaps and bounds since our inception; but we are still in our infancy with a brilliant future ahead of us. Since 1976, and now with 12 editions, *Cohen's Pathways of the Pulp* has always been about the art and science of endodontics, with an emphasis on evidenced-based direction rather than anecdotal guidance. The dedicated contributing authors have generously given their time to meticulously describe what is considered the state of the art of our specialty. We are hopeful that future editions will guide us toward enhanced endodontic outcomes, with the never-ending pursuit of endodontic excellence.

Louis H. Berman
Kenneth M. Hargreaves

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